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Effects of purified herbal extract of *Salvia miltiorrhiza* on ischemic rat myocardium after acute myocardial infarction

Jian Sun[a,1], Shan Hong Huang[b,1], Benny K.-H. Tan[a], Matt Whiteman[b], Yi Chun Zhu[c], Ya Jun Wu[d], Yeekong Ng[d], Wei Duan[b], Yi Zhun Zhu[a,*]

[a]*Department of Pharmacology, National University of Singapore, Singapore*
[b]*Department of Biochemistry, National University of Singapore, Singapore*
[c]*Dept. of Physiology and Pathophysiology, Key Laboratory of Molecular Medicine of the Ministry of Education, Fudan University Shanghai Medical College, Shanghai 200032, China*
[d]*Department of Anatomy, National University of Singapore, Singapore*

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Abstract

In the current study, we compared purified *Salvia miltiorrhiza* extract (PSME) with Angiotensin-converting enzyme inhibitor, Ramipril, in in vitro experiments and also in vivo using animal model of myocardial infarction. PSME was found to have a significantly higher trolox equivalent antioxidant capacity which indicated a great capacity for scavenging free radicals. PSME could also prevent pyrogallo red bleaching and DNA damage.

After 2 weeks treatment with PSME or Ramipril, survival rates of rats with experimental myocardial infarction were marginally increased (68.2% and 71.4%) compared with saline (61.5%). The ratios of infarct size to left ventricular size in both PSME-and Ramipril-treated rats were significantly less than that in the saline-treated group. Activity of cardiac antioxidant enzyme superoxide dismutase (SOD) was significant higher while level of Thiobarbituric acid-reactive substances (TBARs) was lower in the PSME treated group. Purified and standardized Chinese herb could provide an alternative regimen for the prevention of ischemic heart disease.

Keywords: Purified *Salvia miltiorrhiza* extract (PSME); Myocardial infarction; *Salvia miltiorrhiza*; Antioxidant; Infarct size; Ramipril

* Corresponding author. Department of Pharmacology, Faculty of Medicine, National University of Singapore, 18 Medical Drive 117597, Singapore. Tel.: +65 6 874 3676; fax: +65 6 873 7690.
E-mail address: phczhuyz@nus.edu.sg (Y.Z. Zhu).
[1] Contributed equally.

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Introduction

Ischaemic heart diseases, especially acute myocardial infarction (MI), remain the leading cause of death in both developed and developing countries as seen over the past quarter century (Mochizuki et al., 1998; Zhu et al., 1998). Reduction of mortality rate and prevention of myocardial infarction are of utmost importance. Western drugs such as angiotensin-converting enzyme (ACE) inhibitors, calcium channel blockers, angiotensin II receptor antagonists, etc. have been proven to have cardioprotective effects in both preclinical and clinical studies. However, there is a growing interest in the use of alternative medicine for long-term prevention of heart attack in high risk patients.

Salvia miltiorrhiza (SM) is an important Chinese natural herb used in the treatment of many diseases, especially ischemic cardiovascular diseases (Zhu and Zhu, 2002; Ji et al., 2003; Zhu et al., 2004; Ji et al., 2004). It is considered to possess 'slightly cold' and 'bitter' properties and enters the 'heart', 'pericardium', and 'liver' channels. Experiments using modern techniques showed that SM dilates coronary arteries, increases coronary blood flow and scavenges free radicals in patients with ischemic diseases, thus reducing cellular damage from ischemia (Anversa et al., 1993). Clinical trials have also indicated that SM is an effective medicine for angina pectoris, myocardial infarction and stroke (Zhu and Zhu, 2002; Ji et al., 2003; Zhu et al., 2004; Ji et al., 2004; Anversa et al., 1993 and Ji et al., 2000) To date, 20 major ingredients in SM have been identified (Ji et al., 2000). Purified SM used in the current study is an extract consisting of six active water-soluble ingredients from the 20 major ingredients of rude SM.

ACE inhibitors have been gradually introduced into the treatment of hypertension, congestive heart failure and myocardial infarction since the 1970s (Zhu et al., 1998; Linz et al., 1995; Zhu et al., 1996; Zhu et al., 1997). Experimental studies also showed that ACE inhibitors administered chronically before acute MI might limit myocardial infarct size, improve cardiac function and prevent cardiac hypertrophy (Zhu et al., 1998 and Stauss et al., 1994).

In our previous study, we demonstrated that SM reduced mortality and infarct size after MI in rats (Ji et al., 2003) SM was found to reduce oxidant stress from MI by increasing the activities of hepatic-antioxidant enzymes. In the present study, the effects of purified SM in comparison with Ramipril, a well-known ACE inhibitor, on ischemic myocardium in rats after experimental induced acute MI were investigated.

Materials and methods

Animal model

Healthy male adult Wistar rats (4–6 months old) weighing from 230 to 250 g were obtained from the Laboratory Animal Center, National University of Singapore. The experiments were performed according to internationally accepted guidelines for care and use of laboratory animals. Myocardial infarction was induced by ligation of the left coronary artery (close to its origin from the aorta) of rats using a modified version (Stauss et al., 1994; Zhu et al., 2000) of the technique originally described by Selye (Selye et al., 1960) The current work has been received approval from the local ethics committee.

Purified SM

Purified *Salvia miltiorrhiza* extract (PSME, also known as Angino®) was from CMM Corporation Pte Ltd. (Singapore, batch No. IV-S-900-2, 30-4-2003) which was originally developed by Shanghai Material Medica Bioengineering Institute. It is a water soluble extract from *Salvia miltiorrhiza* which contains 6 active bioactive ingredients: Salvianolic acid A, Salvianolic acid B, Rosmarinic acid, 3,4-dihydroxyphenyl lactate (Danshensu), Protocatechualdehyde and Caffeic acid [Technical data offered by Shanghai Materia Medica Bioengineering Institute and CMM Corporation Pte. Ltd., Fig. 1]. Beside these 6 active water soluble ingredients, crude *Salvia miltiorrhiza* extract is mixed with other lipophilic ingredients like Tanshinone I, Tanshinone IIa and IIb, Cryptotanshinone, Isotanshinone I, Isotanshinone II, Isocryptotanshinone, Tanshinol I, Tanshinol II and Vitamine E etc. (Zhu et al., 2004; Ji et al., 2004; Anservsa et al., 1993; Ji et al., 2000).

In vitro experiment

Inhibition of ABTS$^+$ assay

2,2′-azinobis(3-ethylbenothiazoline-6-sulfonic acid) (ABTS) can be quantified spectrophotometrically at 734 nm. To examine the antioxidant effects of PSME, 10 μl or 20 μl of 100 μg/ml PSME were mixed with 1 ml of ABTS reagent and the absorbance was measured at 734 nm after 3 min incubation at room temperature. Trolox was used as a standard and the capacity of free-radical scavenging was expressed as percentage trolox equivalent antioxidant capacity (TEAC) as described previously (Ji et al., 2003; Long and Halliwell, 2001).

Inhibition of pyrogallol red bleaching by ONOO$^-$/HOCl (PR test)

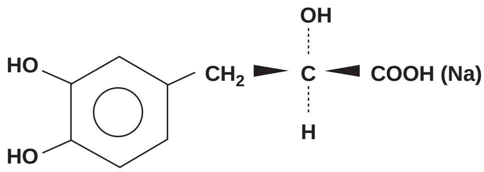
Pyrogallol Red Bleaching by Hypochlorous acid (HOCl) and Peroxynitrite ($ONOO^-$) Assay Pyrogallol red (pyrogallosulphonephthalein; 100 μM) was dissolved in 100 mM K_2HPO_4-KH_2PO_4 buffer, pH 7.4 and various antioxidants, and left at room temperature for 10 min. After this time, 125 μM HOCl or 200 μM $ONOO^-$ were added and the sample were diluted 1:4 before analysis by UV-visible spectroscopy. The decreases in absorbance at 542 nm were recorded as previously described (Long and Halliwell; 2001).

Evaluation of DNA damage using GC/MS

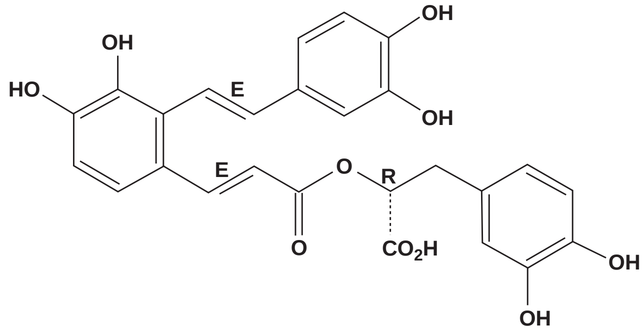
100 μg calf thymus DNA samples were pre-incubated with PSME at various concentrations. HOCl (250 μM) was added to the DNA reaction mixture, followed by incubation at 37 °C for 1 hour. DNA samples were then dialyzed against water for 22 hours.

Hydrolysis was done by addition of 0.5 ml 60% v/v cold formic acid and heating at 150 °C for 45 min in an evacuated, sealed hydrolysis tube after addition of respective internal standards. Samples were cooled and lyophilized.

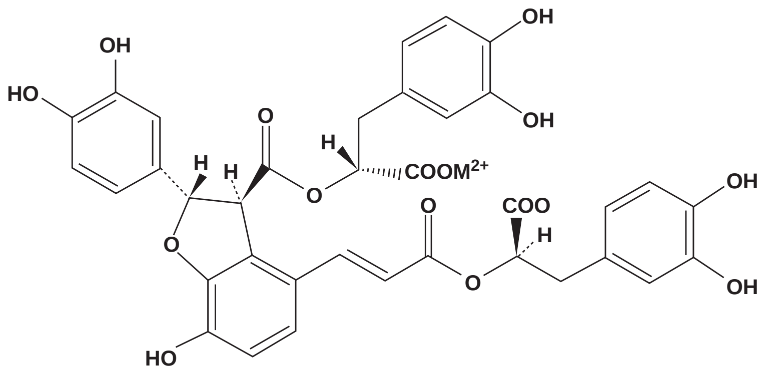
After acid hydrolysis of DNA, samples and calibration standards were derivatised in poly(tetrafluoroethylene)-capped glass vials after purging with nitrogen. 75 μl of a BSTFA (+1% TMCS)/acetonitrile/ethanethiol (16:3:1 v/v) mixture was added and mixed well at 23 °C for 2 hrs. Oxidized DNA was analyzed by GC/MS (Agilent 6890 gas chromatograph and interfaced with an Agilent MSD 5973) as we previously described (Ji et al., 2004; Whiteman et al., 2002a,b).



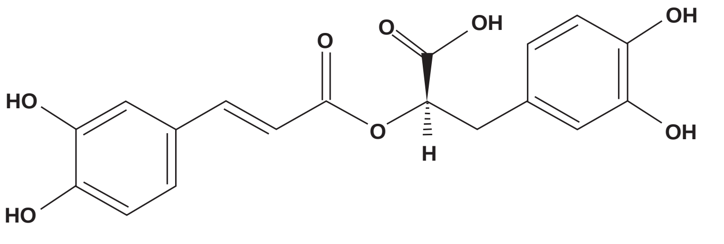
3,4-dihydroxyphenyl lactate



Salvianolic acid A



Salvianolic acid B



Rosmarinic acid

Fig. 1. Stereochemical structures of 4 active bioactive ingredients purified from *Salvia miltiorrhiza*.

In vivo experiment

A total of 98 rats were housed under diurnal lighting conditions and allowed standard rat chow and tap water *ad libitum*. Rats were randomly allocated into four groups: 12 rats given water (4 ml/kg/day) (sham group); 26 rats in AMI group given water as the vehicle (4 ml/kg/day) (AMI group); 22 rats in AMI-PSME group given PSME (100 mg/kg/day) (AMI-PSME group), and 38 rats in AMI-R group given Ramipril (1 mg/kg/day) (AMI-R group). Rats in the AMI, AMI-PSME and AMI-R groups were induced with AMI by ligation of the left anterior coronary artery after one week of oral (gavage) administration. Rats in the sham group were subject to the same procedure except ligation. Surviving rats were sacrificed 1 day after surgery.

Morphological examinations of rat heart with AMI

5–6 hearts in AMI, AMI-PSME, AMI-R groups were stained by Tetrazolium-Blue and kept in 4% phosphate-buffered paraformaldehyde for morphological examination. The sizes of the left ventricle and the infarct area were calculated by Scion Image software as done previously (Ji et al., 2003).

Cardiac antioxidant activity assays

SOD activity was assayed by an improved method of Marklund (Ji et al., 2003; Ji et al., 2004). Briefly, 1 mL cuvette contained 100 μL of 1 mol/L Tris-HCl-5 mmol/L EDTA, pH 8.2, and varying amounts of the enzyme sample (supernatant A) ranging from none in the control to 50 μL to provide more than 50% inhibition. Distilled H_2O was added up to 980 μL. After 10 mins of incubation at room temperature, 20 μL of a 10 mmol/L pyrogallol solution prepared in 10 mmol/L HCl was added. The increase in optical density (O.D.) was measured at 420 nm with the spectrophotometer. The amount of enzymes inhibiting 50% superoxide-dependent oxidation of 0.2 mmol/L pyrogallol is defined as one unit of SOD activity.

GPx activity was assayed by the amended method of Beutler (Ji et al., 2003; Ji et al., 2004). A reaction mixture was made up with 100 μL 1 mol/L Tris-HCL buffer (5 mmol/L EDTA, pH 8.0), 20 μL GSH (100 mmol/L), 100 μL Glutathione reductase (10 U/mL), 100 μL NADPH (2 mmol/L), 5 μL supernatant B and 575 μL distilled H_2O with 1.75 mmol/L sodium azide to eliminate the interference of catalase activity. After a 10-min incubation period at room temperature, 100μL t-Butyl hydroperoxide was added to initiate the reaction. The rate of disappearance of reduced NADPH was measured on the spectrophotometer at 340 nm (ϵ = 6.22 $mol^{-1}cm^{-1}$). The data was expressed as nmol of NADPH oxidized to NADP per min per mg protein.

Catalase was assayed by a method modified from Beutler (Ji et al., 2003; Ji et al., 2004). Briefly, 1 mL of supernatant A and 0.01 mL ethanol was incubated on ice for 30 min. 20 μL incubated supernatant was then taken out and added to a 980 μL assay mixture containing 50 μL of 1 mol/L Tris-HCl EDTA buffer (5 mmol/L EDTA, pH 8.0), 900 μL of 10 mmol/L H_2O_2 and 30μL distilled H_2O. The rate of decomposition of H_2O_2 was monitored for 1 min (ϵ = 0.071 $mol^{-1}cm^{-1}$) from changes in the O.D. at 240 nm. The specific activity was expressed as μmol H_2O_2/min/mg protein.

Thiobarbituric acid-reactive substances (TBARs) assays in the left ventricle

Heating tissues with thiobarbituric acid (TBA) in an acidic medium gives rise to the formation of a red 1:2 malonaldehyde: TBA adduct with absorption maximum at 532 nm. Here we used the method as we previously reported (Ji et al., 2003; Ji et al., 2004). Briefly, 0.2 mL of the homogenate which was set aside after spinning at 300 g for 10 min was used. 0.2 mL of 8.1% dodecyl sulfate sodium salt (SDS), 1.5 mL of 20% acetic acid (pH 3.5 adjusted by 10 N of NaOH), 0.02 μL of 2% BHT and 0.08 mL distilled water were added in turn. After spinning at 10,000 g for 15 min, 1.5 mL of supernatant were transferred into glass tubes followed by addition of 1.5 mL of 0.8% TBA. The mixture was heated at 95 °C for 30 min in a boiling water-bath. After the reaction, the mixture was cooled in an ice-bath; the cold TBA reactants were measured at 532 nm after centrifugation at 5,000 rpm for 5 min. The TBARs are the secondary products of lipid peroxidation. These secondary products are mainly aldehydes, the major compound being malondialdehyde (MDA). The TBARS values thus obtained were expressed as nmol of MDA per gram protein.

Statistical analysis

One-way ANOVA with post hoc Bonferroni multiple comparisons were performed to compare the differences of the infarct areas and biochemical assays among groups. All values were expressed as mean ± SD. Comparisons with $p < 0.05$ were considered statistically significant.

Results

In vitro experiment

Fig. 2 illustrates the effects of ascorbic acid, PSME and Ramipril (all at the same final concentration of 2 μg/mL) reacting with $ABTS^{+}$ at 734 nm. 1 mM trolox was taken as the standard and calculated in terms of the trolox equivalent antioxidant activity (Whiteman et al., 2001). The result shows that the reduction of the radical cation as the inhibition of absorbance at 734 nm, PSME has a significantly higher TEAC value than Ramipril, and almost the same value as ascorbic acid.

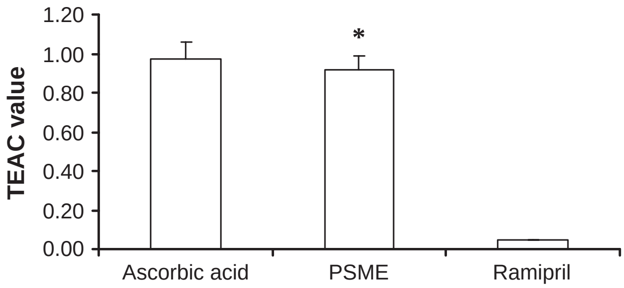


Fig. 2. In vitro effects of ascorbic acid, PSME and Ramipril on TEAC values (ABTS test). Results are expressed as mean±s.d. of at least three independent experiments performed in triplicate. PSME has a significantly higher TEAC value than Ramipril, and almost the same value as ascorbic acid (*p < 0.05 PSME *vs*. Ramipril).

Inhibition of pyrogallol red (PR) bleaching

Fig. 3 shows the inhibition of PR by HOCl (125 μM) or $ONOO^-$(200 μM) using ascorbic acid, PSME and Ramipril. Compared to Ramipril, PSME inhibited PR bleaching by 57% with HOCl and 53% with $ONOO^-$.

Protection against HOCl-mediated DNA damage

PSME significantly prevented DNA damage caused by HOCL. Incubation of DNA with HOCl (250 μM) for 1 hour at 37 °C led to increased formation of DNA base products (Table 1). However, when DNA was pre-incubated with PSME at different concentrations (10, 50,100, 200, 500 and 1000 μg/ml), DNA base products formation was observed to be decreased significantly. It was more significant especially with the higher concentrations as shown in Table 1.

In vivo experiment

The survival rates after 1 day of surgery in sham, AMI, AMI-PSME and AMI-R groups were 100%, 61.5%, 68.2% and 71.4%, respectively (Fig. 4). Rats in AMI-PSME and AMI-R groups had higher survival rates than rats in the AMI group, but the difference was not statistically significant.

The infarct size in each group was shown as a proportion of left ventricular size to the size of the infarcted area in percentage terms in Fig. 4. The ratios of infarct size in both AMI-PSME group and AMI-R group were significantly smaller than that in AMI group, $p < 0.01$ and $p < 0.05$ respectively. But no statistical significance was shown between AMI-PSME group and AMI-R group.

The TBARs level in the sham group was significantly lower than those in the other three groups ($p < 0.001$) as shown in Fig. 5. The TBARs level in AMI-PSME group was significantly lower than both AMI ($p < 0.01$) and AMI-R groups ($p < 0.01$). However, no statistical significance was observed when compared with the difference of TBARs levels between the AMI and AMI-R groups.

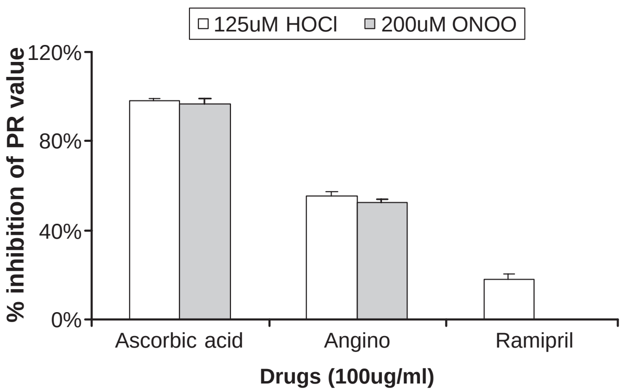


Fig. 3. In vitro effects of ascorbic acid, PSME and Ramipril on inhibition of Pyrogallol Red bleaching by HOCl or $ONOO^-$. Results shown are mean ± s.d. of at least 2 independent experiments performed in duplicate. Compared to Ramipril, PSME inhibited PR bleaching by 57% with HOCl and 53% with $ONOO^-$(*p < 0.05 PSME *vs*. Ramipril).

Table 1
DNA damage analysis from blood samples treated with PSME

Oxidised base	DNA alone	DNA + HOCl	DNA + HOCl + PSME					
			10 μg/ml	50 μg/ml	100 μg/ml	200 μg/ml	500 μg/ml	1000 μg/ml
5-OH, Me Hydantoin	0.03 ± 0.00	0.07 ± 0.01	0.12 ± 0.00	0.14 ± 0.02	0.16 ± 0.00	0.10 ± 0.01	0.12 ± 0.02	0.06 ± 0.01
5-Formyl Uracil	0.00 ± 0.00	0.15 ± 0.01	0.8 ± 0.01	0.05 ± 0.05*	0.01 ± 0.00*	0.01 ± 0.01*	0.00 ± 0.00*	0.00 ± 0.00*
5-OH Uracil	0.02 ± 0.00	1.5 ± 0.14	0.83 ± 0.08*	0.58 ± 0.34*	0.21 ± 0.02*	0.06 ± 0.01*	0.03 ± 0.00*	0.01 ± 0.00*
5-(OH, Me) Uracil	0.13 ± 0.02	0.27 ± 0.04	0.21 ± 0.01*	0.22 ± 0.05*	0.28 ± 0.03	0.16 ± 0.00*	0.12 ± 0.02*	0.09 ± 0.01*
5-OH Cytosine	0.12 ± 0.01	3.65 ± 0.36	1.36 ± 0.08*	1.0 ± 0.44*	0.38 ± 0.10*	0.23 ± 0.01*	0.21 ± 0.05*	0.11 ± 0.09*
Thymine Glycol	0.09 ± 0.01	7.3 ± 0.35	3.21 ± 0.22*	2.19 ± 1.56*	0.32 ± 0.11*	0.14 ± 0.03*	0.09 ± 0.01*	0.07 ± 0.01*
FAPy Adenine	0.49 ± 0.04	0.44 ± 0.02	0.36 ± 0.14	0.36 ± 0.07	0.43 ± 0.09	0.38 ± 0.04	0.34 ± 0.12*	0.31 ± 0.10*
8-OH Adenine	0.02 ± 0.00	0.87 ± 0.06	0.49 ± 0.06*	0.41 ± 0.20*	0.18 ± 0.10*	0.17 ± 0.09*	0.24 ± 0.06*	0.07 ± 0.02*
2-OH Adenine	0.59 ± 0.03	0.66 ± 0.04	0.23 ± 0.02*	0.20 ± 0.06*	0.25 ± 0.07*	0.20 ± 0.05*	0.20 ± 0.07*	0.09 ± 0.04*
FAPy Guanine	0.14 ± 0.03	0.51 ± 0.11	0.35 ± 0.30*	0.08 ± 0.00*	0.10 ± 0.01*	0.06 ± 0.00*	0.05 ± 0.02*	0.02 ± 0.01*
8-OH Guanine	0.17 ± 0.01	0.17 ± 0.02	0.18 ± 0.04	0.15 ± 0.04	0.17 ± 0.02	0.14 ± 0.05	0.10 ± 0.01*	0.09 ± 0.04*

Dose response inhibition of HOCl reduced DNA damage by PSME. DNA (1.0 mg/ml) was oxidized using HOCl (250 μM) for 1 hour at room temperature. The samples were then dialyzed against water for 22 hours, freeze dried and hydrolyzed with 60% (v/v) formic acid. Oxidized DNA base products were analyzed by GC/MS after derivatization for at least 2 hours. 'DNA alone' was a negative control and 'DNA plus HOCl' served as positive control. Results are expressed as mean ± s.d. of 3 determinations performed in triplicate and expressed as nmol/mg DNA (*$p < 0.05$ PSME treated groups *vs*. untreated group [positive control]).

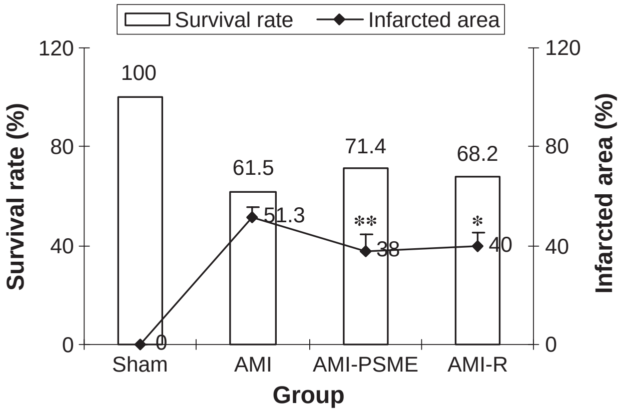


Fig. 4. Survival rates of sham, AMI, AMI-PSME and AMI-R rat groups after MI at the end point of experiment and the ratios of infarct size to left ventricular size in AMI, AMI-PSME and AMI-R groups. AMI: acute myocardial infarction. AMI-PSME: acute myocardial infarction treated with PSME. AMI-R: acute myocardial infarction treated with Ramipril. Values are expressed as mean ± SD (n=5–6). *p < 0.05 AMI group vs. AMI-R group ** p < 0.01 AMI group vs. AMI-PSME group.

The level of SOD activity in sham group was significantly higher than those in the other three groups ($p < 0.001$) (Fig. 5). Only the AMI-PSME group showed significantly higher level of SOD activity compared with the AMI group ($p < 0.05$).

No statistical significance was observed when comparing the differences of GPx and CAT activities among the groups after AMI.

In a morphological study investigating whether long term treatment (3 months) of PSME would cause any cellular toxicity, observation from electro-microscope showed no significant change from cell-structure and morphology.

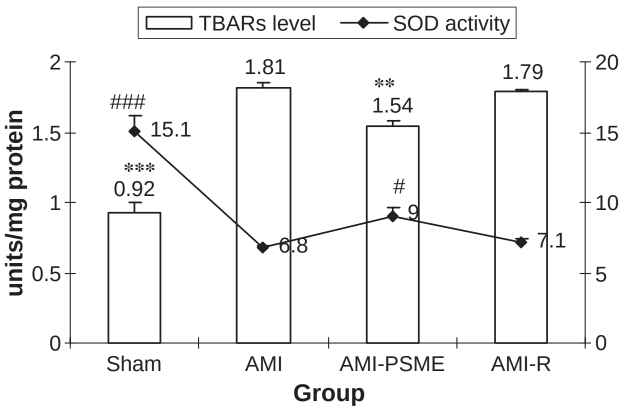


Fig. 5. TBARs levels and cardiac SOD activity in sham, AMI, AMI-PSME and AMI-R groups after myocardial infarction. Values are expressed as mean ± SD (n=3–4). **p < 0.01 AMI-PSME group vs. AMI, or AMI-R groups; *** p < 0.001 sham group vs. AMI, AMI-PSME, or AMI-R groups. #p < 0.05 AMI-PSME group vs. AMI group; ### p < 0.001 sham group vs. AMI, AMI-PSME, or AMI-R groups. AMI: acute myocardial infarction. AMI-PSME: acute myocardial infarction treated with PSME. AMI-R: acute myocardial infarction treated with Ramipril.

Discussion

The results in this study demonstrated for the first time that PSME had similar cardioprotective effects as Ramipril, an ACE inhibitor, producing a higher survival rate and causing smaller infarct size compared to vehicle. Moreover, PSME showed antioxidant effects including higher TEAC value, greater inhibition of PR bleaching, prevention of DNA damage, inhibition of lipid peroxidation and preservation of SOD activity. But it has no direct effect on lowering blood pressure, unlike Ramipril.

Both survival rate and infarct size are important parameters for evaluating the effectiveness of cardiovascular drugs in the treatment of ischemic heart diseases. Although probably due to a small sample size, we failed to get a statistical significance in terms of survival rate, we found that rats treated with either PSME or Ramipril tended to have longer survival after MI (marginal increase of the survival rate in PSME treated group compared to vehicle group). Such results are in accordance with our previous study when the effects of Ramipril were compared to SM injection in a similar animal model (Ji et al., 2003; Ji et al., 2004). Both PSME (100 mg/kg/day) and Ramipril (1 mg/kg/day) significantly decreased the infarct size of the left ventricle after AMI in rats. It is well-known that SM could dilate coronary arteries, increase coronary blood flow and scavenge free radicals in ischemic diseases, thus reducing cellular damage from ischemia. PSME, formed by SM's active ingredients, might have similar mechanisms to SM.

Oxidative stress is one of the major concerns in the treatment of ischemic heart diseases. There is ample evidence for a detrimental role of ROS in cardiovascular disease (Dhalla et al., 2000; Sawyer et al., 2002) . PSME showed significant effects of reducing radical cation and DNA damage in vitro. The phenomenon in our in vivo experiment that activities of antioxidant enzymes which decreased significantly within the first day of surgery coincided with the finding of elevated TBARs level strongly argues for the presence of increased free radical activity as well as of the reciprocal decreased endogenous antioxidant activity. Rats treated with PSME had high antioxidant activities but low MDA production in comparison with rats treated with water or Ramipril, which implies that PSME may affect the level of endogenous antioxidants or oxidative stress or both. A recent study reported that Ramipril is also able to decrease endogenous antioxidant activity in rats with myocardial infarction (Linz et al. 2003). In a recent chronic study, we found that 3 months of PSME treatment significantly increased enzyme-like particles on electron-microscopy in the livers of these animals compared to saline treated-groups (Fig. 6). A further study will be carried out to characterize the anti-oxidant enzymes.

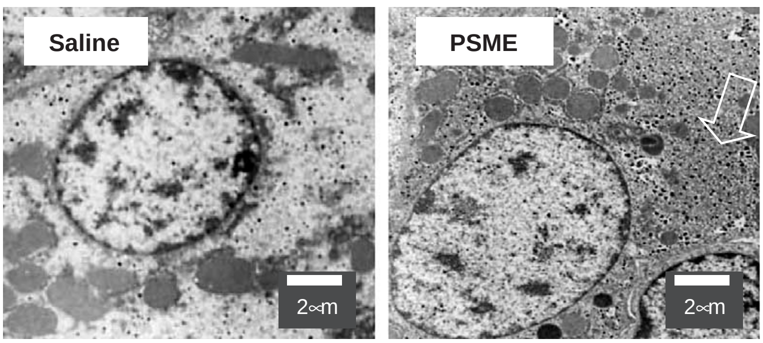


Fig. 6. Electron-microscopy of livers from PSME treated animals. Increased enzyme-like black particles were observed (as indicated by arrows).

Several active ingredients in SM, such as 3,4-dihydroxyphenyl lactate and tanshinone, could scavenge superoxide anions generated from the xanthine-xanthine oxidase system or the lipid free radicals generated from lipid peroxidation of the myocardial mitochondrial membranes (Anversa et al., 1993). Our previous study also demonstrated that injection of SM increased the activities of hepatic antioxidant enzymes, including SOD, CAT and GPx, and inhibited hepatic and myocardial lipid peroxidation in AMI rats (Ji et al., 2003). Therefore, some of the ingredients in PSME might exert similar beneficial effects as shown above.

Acknowledgement

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